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- 5 Food and Drug Administration (FDA) (2018). *Bioanalytical Method Validation Guidance for Industry*. Washington, DC: Office of Communications, Division of Drug Information Center for Drug Evaluation and Research.
- 6 Council Directive 96/23/EC of 29 April 1996 on measures to monitor certain substances and residues thereof in live animals and animal products and repealing Directives 85/358/EEC and 86/469/EEC and Decisions 89/187/EEC and 91/664/EEC.
- 7 2000/657/EC: Commission Decision of 16 October 2000 adopting Community import decisions for certain chemicals pursuant to Council Regulation (EEC) No 2455/92 concerning the export and import of certain dangerous chemicals.
- 8 Standard ISO 11843-2 (2000). *Capability of Detection — Part 2: Methodology in the Linear Calibration Case*. Genève: ISO.
- 9 Commission Regulation (EU) 2019/1871 of 7 November 2019 on reference points for action for non-allowed pharmacologically active substances present in food of animal origin and repealing Decision 2005/34/EC.
- 10 Commission Implementing Regulation (EU) 2021/808 of 22 March 2021 on the performance of analytical methods for residues of pharmacologically active substances used in food-producing animals and on the interpretation of results as well as on the methods to be used for sampling and repealing Decisions 2002/657/EC and 98/179/EC.
- 11 Commission Regulation (EU) No 37/2010 of 22 December 2009 on pharmacologically active substances and their classification regarding maximum residue limits in foodstuffs of animal origin.
- 12 Commission Regulation (EC) No 124/2009 of 10 February 2009 setting maximum levels for the presence of coccidiostats or histomonostats in food resulting from the unavoidable carry-over of these substances in non-target feed.

10

Examples of MU Application

10.1 Standard Addition Method and Drug Quality

Access to appropriate, often patent-protected medicines, can be extremely difficult for low- and middle-income countries, and, in some cases, for high-income countries. For the latter, vulnerable populations, such as prisoners, drug addicts, or migrants, who have limited access to certain therapies, could be affected. In addition to its ethical aspects, this issue raises public health problems for controlling certain pandemics, such as acquired immuno deficiency syndrome (AIDS).

The price of original patented medicines appears to be a limiting factor for widespread treatment. Indeed, many indigent patients are treated by public health institutions whose resources are limited and which therefore have difficulty fulfilling their missions. For this reason, in several countries, including the United States, Canada, and European nations such as Switzerland, nonprofit organizations known as buyers' clubs have been set up to facilitate access to these essential medicines for vulnerable populations.

These buyers' clubs use a "right to import unlicensed ready-to-use medicines from individuals" defined in international agreements [1]. Even if buyers' clubs select pre-qualified manufacturers for import, the possibility of receiving drug products of insufficient quality cannot be excluded due to the high frequency of falsified molecules coming from exporting countries.

Therefore, a quality control test upon arrival of the drug could help to ensure patient safety. However, this control must be carried out, differently from the classical control established in the pharmaceutical industries, because there are two main constraints to developing a satisfactory assay method:

- The first is the lack of knowledge of the exact nominal content of the drug and the impossibility of having validation materials with guaranteed content.
- The second is the possible presence of matrix interferences that can bias the measurements.

In addition, external calibration for quantification is made difficult by the small number of samples available and the large potential differences in composition for a drug containing the same active principal ingredient (API). To overcome these obstacles, the control of various antiretroviral drugs received by buyers' clubs could

be based on standard addition method (SAM). The pros and cons of this quantification mode are extensively described in Section 1.4.1.

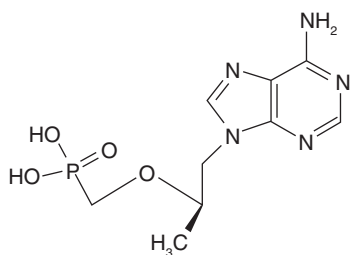
It is to be remembered that SAM is generally said to avoid matrix effects and simultaneously corrects calibration biases due to diverse sources, such as medicine coating, excipients, or interfering matrix components. It is also extremely useful when it is not possible to have surrogate validation materials with content which is exactly known.

There are various SAM operating procedures, and the chosen one consists in adding increasing amounts of a standard solution of the analyte to be determined in constant aliquots of the sample. In this case, the concentrations of all the matrix components are kept constant, except for the molecule to be determined, here being the API. Nevertheless, since the sample predicted concentration is calculated by extrapolating a response function, this method may induce a bias – generally small – due to possible multiplicative effect interferences.

The procedure, although very efficient in many cases, is often criticized from a statistical point of view because extrapolation is prone to increase prediction uncertainty, especially when the amount of spike is not properly adjusted with respect to the initial signal measured in the tested sample.

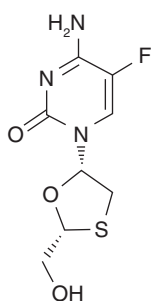
When it is possible to make several simultaneous measurements, typically at different wavelengths, the technique called H-point standard addition method (HPSAM) dramatically reduces the risk of not detecting interferences. In the original version developed for a spectrophotometry-based method, it consisted of multiple determinations of an analyte by SAM at several wavelengths corresponding as much as possible to different signals varying with the analyte concentration and constant for the interferent [2]. It is then possible to simultaneously compute several in-sample calibration curves.

Although they have different slopes because the sensitivity may vary, in the absence of important additive bias due to matrix effects, they must intersect with the concentration axis at close points and give close extrapolated concentrations. If the additive bias is critical, a modified HPSAM was proposed, including chemical modifiers [3].



Tenofovir (TDF)

The purpose of this example is to show how to estimate the measurement uncertainty (MU) when applying the H-point operating procedure. It consists in the determination of two antiretroviral active ingredients present in various pharmaceutical formulations used for the treatment of AIDS, namely tenofovir (TDF) and emtricitabine (FTC).



Emtricitabine (FTC)

Two approaches have been used to estimate the MU. The first one uses the classical statistical information available from the extrapolated concentration obtained by SAM, the second one is based on a complete experimental design to build an accuracy profile to demonstrate the continuity between the MAP procedure and the estimation of MU, whatever the quantification method.

It should be noted that in many similar cases, complete validation is impractical or even impossible. This type of procedure requires relatively massive quantities of validation materials of exactly known contents prepared on purpose. Moreover, in the case of drugs imported by buyers' clubs, the diversity of formulations would require specific validation for each specialty.

Capillary zone electrophoresis (CZE) is an analytical technique recognized as performing well for the control of adulterated pharmaceutical formulations and/or the quality control of drugs, in general. This technique is, therefore, particularly well suited to the case in hand: it is economical thanks to a reduced solvent consumption of the order of 1 μ l per analysis, and it avoids the use of expensive and environmentally damaging organic solvents.

Quantification by CZE is relatively easy, and many studies have shown that its performance is similar to conventional methods, such as liquid chromatography. In addition, for this study, the instrument is coupled to a diode array UV detector (DAD), which allows the measurement of responses at different wavelengths, as required for the HPSAM, in this case, 200, 210, and 254 nm.

The preparation of the sample is simple: it is ground, then suspended in a solvent, which after dilution, forms the injected solution. For example for a drug advertised at 240 mg of active ingredient per tablet, the dilution factor applied was 1 : 2500 so that it to be around 0.1 mg/ml in the final solution and be in a range where the UV signal is in the linear range of the detector used.

10.1.1 SAM Without Replication

According to the chosen SAM procedure, equal volumes of the sample solution are taken, and known quantities of a standard solution are added separately. In this case, in addition to the original sample, two spikes are done, which correspond to approximately 50% and 100% of the expected signal for the likely nominal content. This is a standard protocol. All these aliquots are then brought to the same final volume and measured by CZE.

More details on this procedure, the reagents used, and the instrumental conditions are available in [4]. In the absence of significant matrix effects, at the chosen

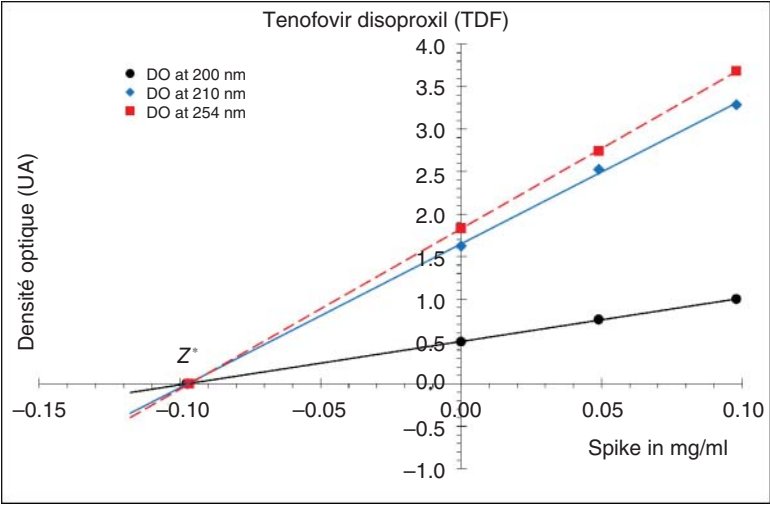


Figure 10.1 Example of assay obtained by SAM on a pharmaceutical product containing TDF. Three estimates of extrapolated concentration can be computed. For sake of clarity only one value is shown.

wavelengths, the three calibration lines, although with different slopes, intersect with the concentration axis for extrapolated concentrations remarkably close to each other. This approach makes an optimized use of the effort required for sample preparation allowing several replicates to be obtained for the same sample.

Figure 10.1 shows an example of an assay performed under the described conditions for a medicine containing TDF at the claimed nominal content of 250 mg/tablet. Measurements were made at three wavelengths, namely 200, 210, and 254 nm, with standard additions at 0.049 and 0.098 mg/ml. As usual, the signal is shown on the vertical axis, expressed in arbitrary units (AU), and the amounts of analyte added on the horizontal axis.

The estimated content in the working sample denoted Z^* (star) is obtained by extrapolating the regression lines to the point where the response is 0. As shown in the figure, it is in the negative part, around 0.1 mg/ml. Since a dilution factor of 1 : 2500 was applied, the concentration per tablet is around 250 mg. For each wavelength, three spiked samples are measured.

Let us note each wavelength with the index j , with $1 \leq j \leq J$ since the measurements, once converted into concentration, will be assimilated to replicates. Each regression line connecting the measurement values is calculated using the same equations and notations as in Section 2.3.1, except for subscripts.

X_n	Spike concentration in mg/ml
$1 \leq n \leq N$	Number of spiked points, in the example $N = 3$
Y_n	Optical density measured in AU
Z^*	Extrapolated sample concentration